DEPENDENCY AWARE PQC MIGRATION FRAMEWORK FOR DIGITAL BANKING

End-to-end system architecture | Graph-aware migration planning and risk-bound banking transactions

1 ASSET INVENTORY & DEPENDENCY MODEL

Synthetic data and service-weighted risk index

Banking inventory
2,100 assets | 60 services
2,862 dependency records
Assets / services / dependencies CSVs

Load & validate
Schema, tiers, IDs, ranges, dependency endpoints
src/data_loader.py

Build directed graph
Internal / third-party / sub-tier
2,856 unique weighted edges
src/graph_builder.py

Evaluate ecosystem risk
Direct risk + strongest dependency paths + service-weighted ER
src/risk_model.py | src/propagation.py

risk context

2 RISK-BASED MIGRATION PLANNER

Budgeted counterfactual planning, not live deployment

Candidate risk impact
Temporarily set vulnerability to 0
Recompute modeled ΔER
src/remediation.py

Priority calculation
Choose largest positive ΔER / cost within remaining budget
src/prioritization.py

Dynamic migration plan
Select asset; update model; rerank
Budget ceilings: 2.5 / 5.0 / 7.5
experiments/final_evaluation.py

Planning outputs
Ranked asset list, migration trace
New modeled ER; comparison CSVs
results/tables | results/figures

policy refresh

3 TRANSACTION-TIME PQC ENFORCEMENT

ML-KEM / ML-DSA with simulated supplier evidence

Risk-bound policy
Standard: ML-KEM-768 / ML-DSA-65
Elevated: ML-KEM-1024 / ML-DSA-87
pqc_layer/darp_pqc.py

Banking gateway + RBAC
Service policy, actor authorization
Epoch & policy-digest binding
pqc_layer/banking_gateway.py

Cryptographic protection
OpenSSL ML-KEM, ML-DSA
HKDF-SHA256 + AES-256-GCM
pqc_layer/crypto_backend.py

Verify and accept
Signature + binding + replay checks
Reject missing / invalid tier evidence
pqc_layer/multitier_assurance.py

Cross-tier evidence — Simulated third-party + sub-tier signed acknowledgements — **enforced at gateway**

4 EVALUATION & ARTIFACTS

Planning and cryptographic tests reported separately

Migration / graph evaluation
Baselines, adapted QARS, budget sweeps, ablation, scalability

Security / performance checks
PQC operations, policy changes, tier-gate tests, latency, envelope size

Interpretation: migration reduces a synthetic modeled-risk index; supplier evidence is issued by simulated providers, not independently attested real vendors.